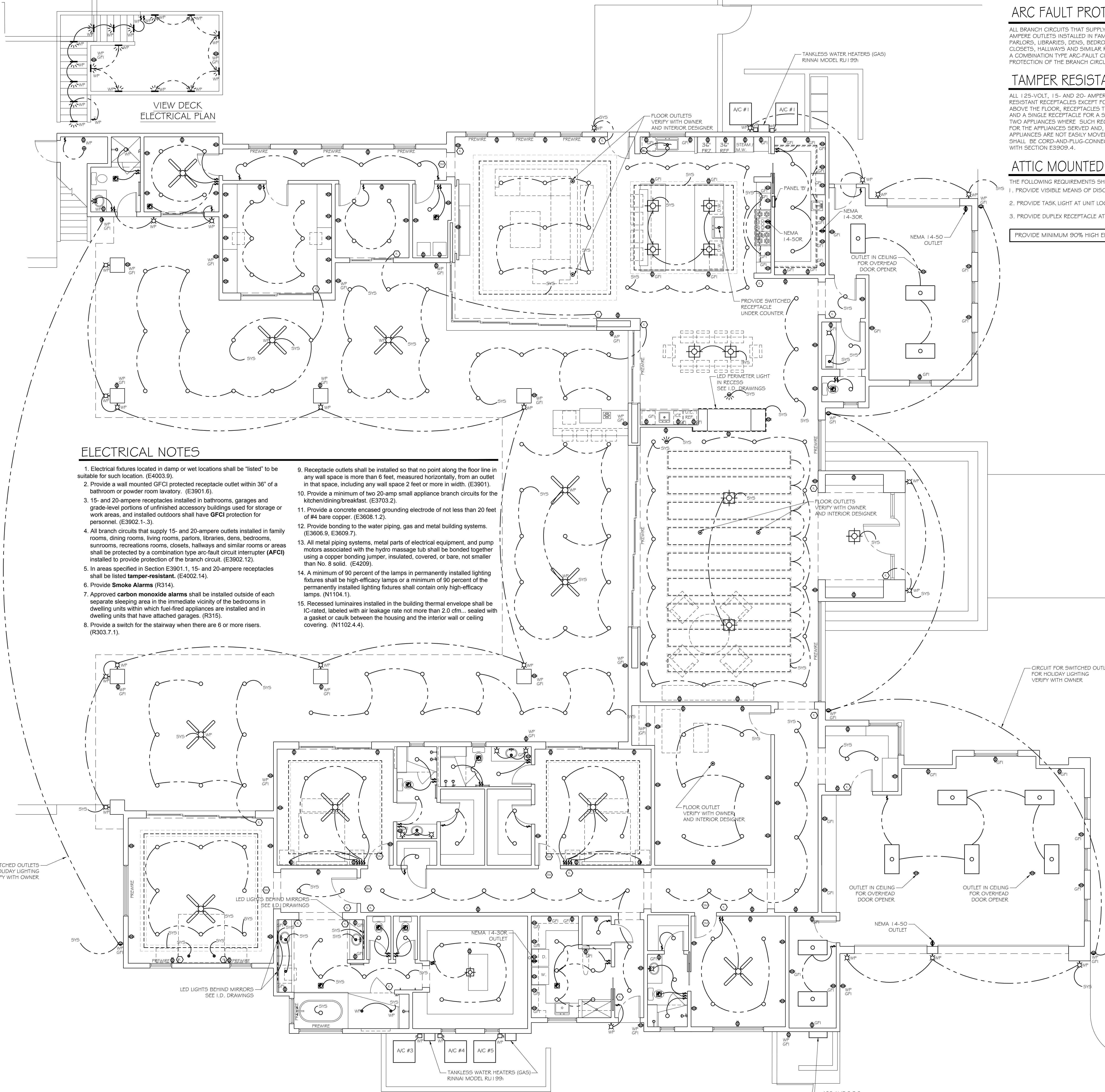




ELECTRICAL NOTES

- 1. Electrical fixtures located in damp or wet locations shall be "listed" to be suitable for such location. (E4003.9).
- 2. Provide a wall mounted GFCI protected receptacle outlet within 36" of a bathroom or powder room lavatory. (E3901.6).
- 3. 15- and 20-ampere receptacles installed in bathrooms, garages and grade-level portions of unfinished accessory buildings used for storage or work areas, and installed outdoors shall have GFCI protection for personnel. (E3902.1-3).
- 4. All branch circuits that supply 15- and 20-ampere outlets installed in family rooms, dining rooms, living rooms, parlors, libraries, dens, bedrooms, sunrooms, recreations rooms, closets, hallways and similar rooms or areas shall be protected by a combination type arc-fault circuit interrupter (AFCI) installed to provide protection of the branch circuit. (E3902.12).
- 5. In areas specified in Section E3901.1, 15- and 20-ampere receptacles shall be listed **tamper-resistant**. (E4002.14).
- 6. Provide **Smoke Alarms** (R314).
- 7. Approved **carbon monoxide alarms** shall be installed outside of each separate sleeping area in the immediate vicinity of the bedrooms in dwelling units within which fuel-fired appliances are installed and in dwelling units that have attached garages. (R315).
- 8. Provide a switch for the stairway when there are 6 or more risers. (R303.7.1).
- 9. Receptacle outlets shall be installed so that no point along the floor line in any wall space is more than 6 feet, measured horizontally, from an outlet in that space, including any wall space 2 feet or more in width. (E3901).
- 10. Provide a minimum of two 20-amp small appliance branch circuits for the kitchen/dining/breakfast. (E3703.2).
- 11. Provide a concrete encased grounding electrode of not less than 20 feet of #4 bare copper. (E3608.1.2).
- 12. Provide bonding to the water piping, gas and metal building systems. (E3608.9, E3609.7).
- 13. All metal piping systems, metal parts of electrical equipment, and pump motors associated with the hydro massage tub shall be bonded together using a copper bonding jumper, insulated, covered, or bare, not smaller than No. 8 solid. (E4209).
- 14. A minimum of 90 percent of the lamps in permanently installed lighting fixtures shall be high-efficacy lamps or a minimum of 90 percent of the permanently installed lighting fixtures shall contain only high-efficacy lamps. (N1104.1).
- 15. Recessed luminaires installed in the building thermal envelope shall be IC-rated, labeled with air leakage rate not more than 2.0 cfm., sealed with a gasket or caulk between the housing and the interior wall or ceiling covering. (N1102.4.4).

ELECTRICAL PLAN

SCALE: 3/16"=1'-0"

ARC FAULT PROTECTION (2018 IRC)

ALL BRANCH CIRCUITS THAT SUPPLY 120-VOLT, SINGLE PHASE, 15- AND 20- AMPERE OUTLETS INSTALLED IN FAMILY ROOMS, DINING ROOMS, LIVING ROOMS, PARLORS, LIBRARIES, DENS, BEDROOMS, SUNROOMS, RECREATION ROOMS, CLOSETS, HALLWAYS AND SIMILAR ROOMS OR AREAS SHALL BE PROTECTED BY A COMBINATION TYPE ARC-FAULT CIRCUIT INTERRUPTER INSTALLED TO PROVIDE PROTECTION OF THE BRANCH CIRCUIT.

TAMPER RESISTANT RECEPTACLES

ALL 125-VOLT, 15- AND 20- AMPERE RECEPTACLES SHALL BE LISTED TAMPER RESISTANT RECEPTACLES EXCEPT FOR RECEPTACLES LOCATED MORE THAN 5.5 FEET ABOVE THE FLOOR, RECEPTACLES THAT ARE PART OF A LUMINAIRE OR APPLIANCE, AND A SINGLE RECEPTACLE FOR A SINGLE APPLIANCE OR A DUPLEX RECEPTACLE FOR TWO APPLIANCES WHERE SUCH RECEPTACLES ARE LOCATED IN SPACES DEDICATED FOR THE APPLIANCES SERVED AND, UNDER CONDITIONS OF NORMAL USE, THE APPLIANCES ARE NOT EASILY MOVED FROM ONE PLACE TO ANOTHER. THE APPLIANCES SHALL BE CORD-AND-PLUG-CONNECTED TO SUCH RECEPTACLES IN ACCORDANCE WITH SECTION E3909.4.

ATTIC MOUNTED MECH. UNITS

THE FOLLOWING REQUIREMENTS SHALL BE MET FOR ALL ATTIC MOUNTED MECH. UNITS

- 1. PROVIDE VISIBLE MEANS OF DISCONNECT PER E4101.5
- 2. PROVIDE TASK LIGHT AT UNIT LOCATION SWITCHED AT ACCESS SCUTTLE.
- 3. PROVIDE DUPLEX RECEPTACLE AT UNIT LOCATION

PROVIDE MINIMUM 90% HIGH EFFICACY LAMPS (N1104.1)

SYMBOL LEGEND

- SURFACE MOUNTED LIGHT FIXTURE
- 6" RECESSED CAN LIGHT FIXTURE
- HALO MODEL H7CATNB HOUSING WITH HALO MODEL 30WAT BAFFLE
- MEETS ASTM-E285 AND IS DESIGNED TO BE IN DIRECT CONTACT WITH INSULATION AND TO MEET RESTRICTED AIR FLOW REQUIREMENTS
- WALLWASH RECESSED CAN LIGHT FIXTURE
- 4" MINI-CAN (LINE VOLTAGE)
- WALL MOUNTED LIGHT FIXTURE
- WALL MOUNTED LIGHT FIXTURE (LOW PROFILE)
- SUSPENDED LIGHT FIXTURE - PENDANT
- RECESSED PATH LIGHT
- SUSPENDED LIGHT FIXTURE - CHANDELIER
- WALL MOUNTED ART LIGHT (OWNER SELECT)
- WALL MOUNTED SECURITY LIGHT (MOTION SENSOR)
- LED TAPE LIGHTING (CEILING COVE / UNDER CABINET)
- LED GARAGE LIGHT FIXTURE - OWNER SELECT
- RECESSED STEP LIGHTING
- CEILING FAN - VERIFY WITH OWNER
- PROVIDE J-BOX RATED FOR CEILING FAN INSTALLATION
- CEILING MOUNTED HEATER
- VERIFY WITH OWNER
- 100 CFM EXHAUST FAN
- 120V DUPLEX RECEPTACLE
- 120V DUPLEX RECEPTACLE - ONE SIDE SWITCHED
- 120V DUPLEX RECEPTACLE - UNDER COUNTER
- 240V 4-PRONG RECEPTACLE
- FLOOR RECEPTACLE
- 120V 4-PRONG RECEPTACLE
- WALL SWITCH - TWO WAY
- WALL SWITCH - MULTI-WAY
- SMOKE / MONOXIDE COMBINATION DETECTOR
- ALARM CONTROL PANEL
- SYSTEM KEYPAD
- TV OUTLET (COAX AND CAT6)
- DOOR BELL
- FUSED DISCONNECT
- DISPOSAL GRINDER MOTOR
- VERIFY POWER & SWITCH WITH MANUF.
- WP
- GFI
- WP
- SYS
- PREWIRE
- INDICATES LOAD ON LIGHTING CONTROL SYSTEM - COORDINATE WITH AV CONTRACTOR
- INDICATES WINDOWS TO BE PREWIRED FOR MOTORIZED SHADES - VERIFY WITH OWNER

LOAD CALCULATIONS

LIVABLE - 7,156 S.F. X 3 WATTS	21,468 W
GARAGE / STORAGE / MECH. - 1,533 S.F. X 1.5 WATTS	2,300 W
SMALL APPLIANCE CIRCUITS	9,000 W
LAUNDRY CIRCUIT	1,500 W
DISHWASHER / DISP.	3,000 W
WATER HEATER	-
DRYER	-
RANGE	-
OVENS	-
CAR CHARGING OUTLET	17,600 W
MICROWAVE	1,200 W
SUB TOTAL	=56,068 W
FIRST 6 KW @ 100%	6,000 W
REMAINDER @ 40%	19,227 W
SUB TOTAL	=27,227 W
POOL EQUIP.	8,800 W
AIR CONDITIONING	5 UNITS x 8,800 W
TOTAL	= 80,027 W
TOTAL / 240 V	AMPS = 333.45

ONE LINE DIAGRAM

ELECTRICAL SERVICE RESERVED SPACE

THE MAIN ELECTRICAL SERVICE PANEL SHALL HAVE A RESERVED SPACE TO ALLOW INSTALLATION OF A DUAL-POLE CIRCUIT BREAKER FOR "FUTURE SOLAR ELECTRIC" INSTALLATION AND SHALL BE LABELED AS SUCH.

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CHANDLER, ARIZONA

FLOOR PLAN - NOTED

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3

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